EDA Credit Risk Analysis

Introduction:-

- Mount Google Drive to access the dataset files
- Upload the dataset file
- Unzipping the dataset files
- Load the dataset
- Data Understanding
- Data Cleaning
- Univariate Analysis
- Segmented Univariate Analysis
- Bivariate Analysis
- Multivariate Analysis
- Correlation Analysis
- Create new feature/Metrices
- Feature Analysis and Selection
- Conclusion

Objective:-

- To analyze a credit risk dataset containing loan application and client information.
- Identify important factors that influence the likelihood of a client defaulting on a loan.
- Perform exploratory data analysis to uncover patterns, data quality issues, and relationships between variables.
- Provide insights and recommendations to improve credit decision-making and reduce risk.

Dataset Overview:-

- Dataset contains two main files:
  - **Application Data:** 307,511 loan applications with 122 features.
  - **Previous Application Data:** 1,670,214 previous loan applications with 37 features.
- Data types include numerical, categorical, and date/time features.
- Contains missing values and requires cleaning before analysis.

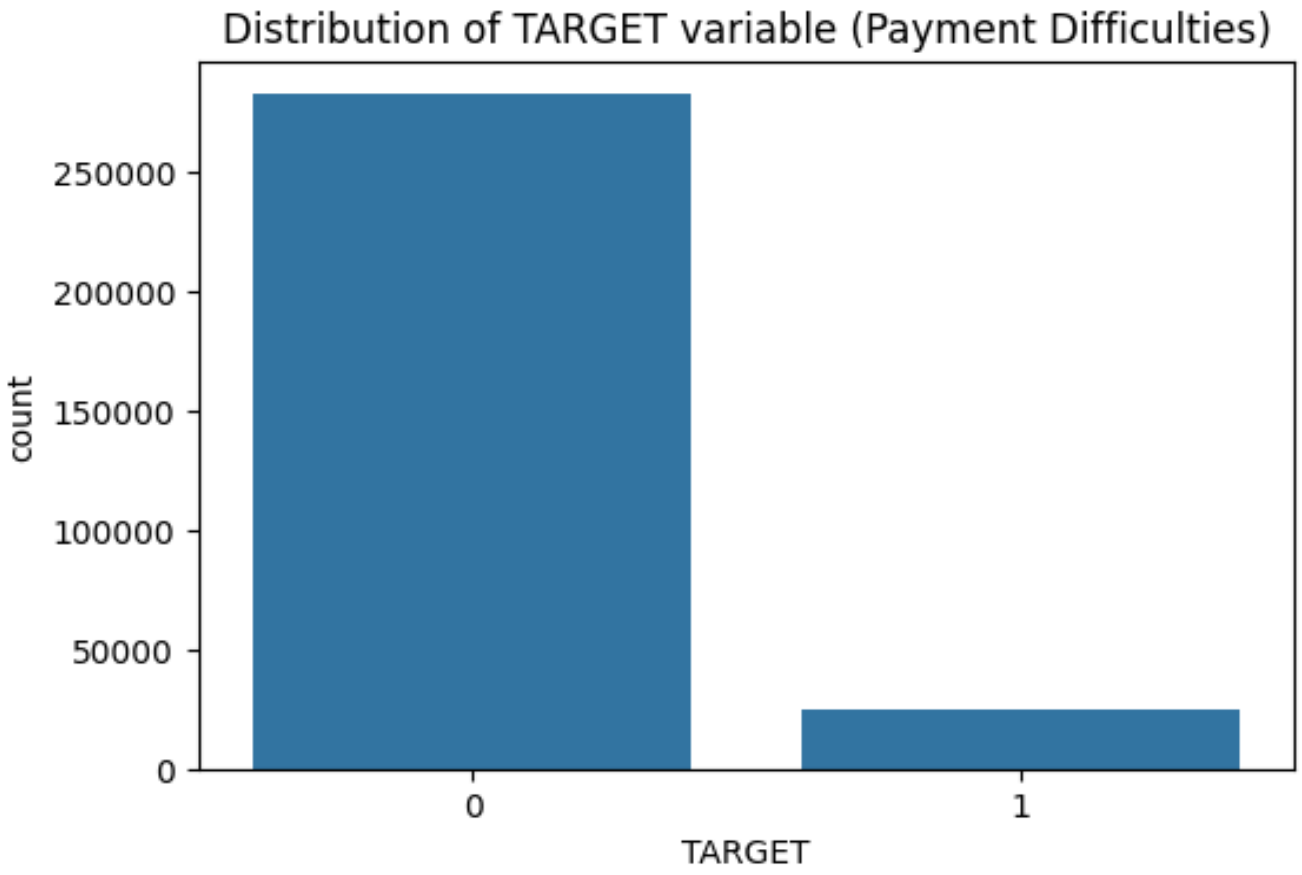
Data Cleaning and Preprocessing:-

- Identified and handled missing values based on threshold (40%).
- Dropped columns with high missing values.
- Imputed missing values:
  - Categorical columns: Mode
  - Numerical columns: Median
- Converted data types where necessary for analysis.

Univariate Analysis:-

•Shows the proportion of clients who repaid their loans successfully vs. those who faced payment difficulties.

•Understanding this balance helps in building and evaluating predictive models.

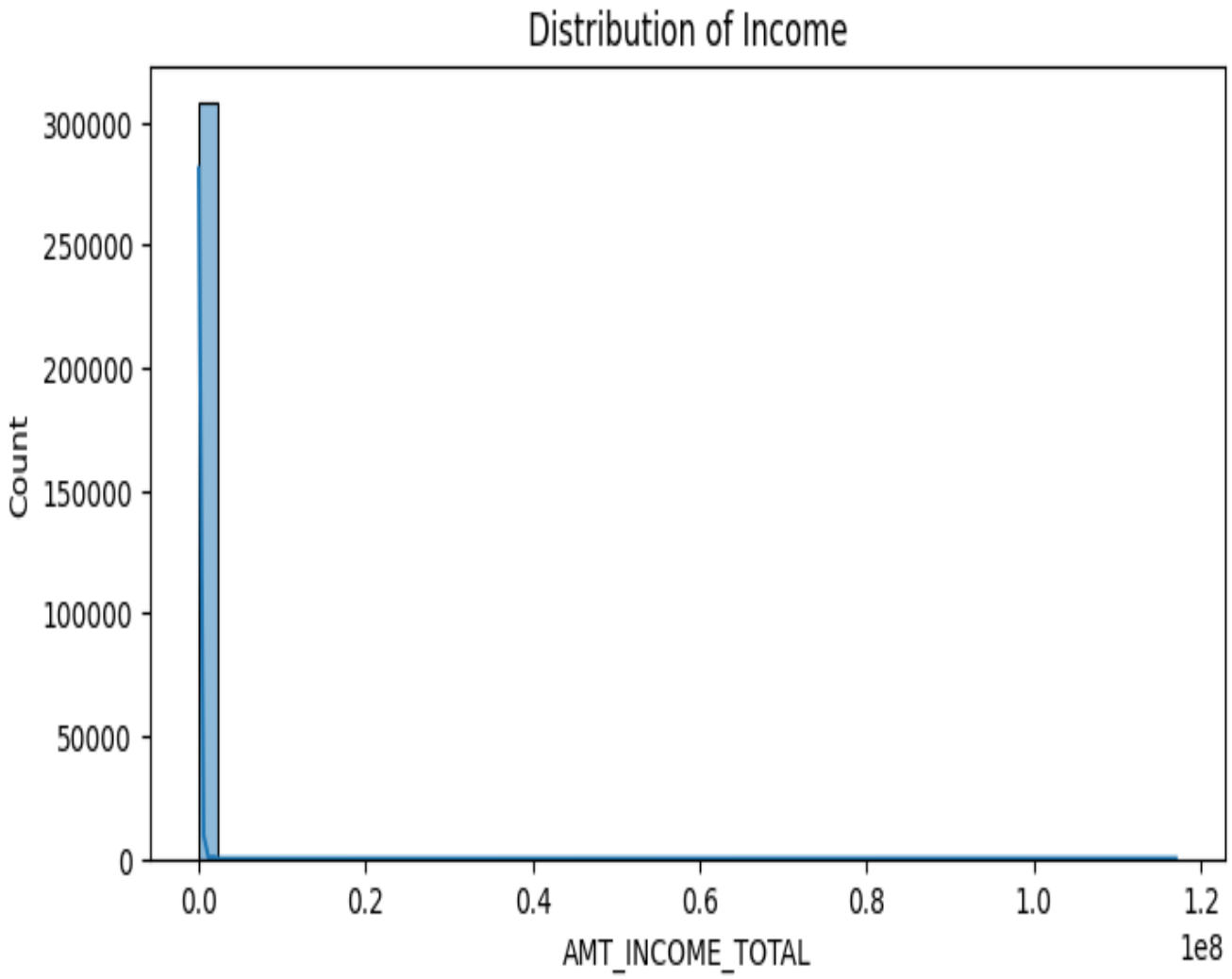


Univariate Aanlysis:-

•Displays how client incomes are spread across the dataset.

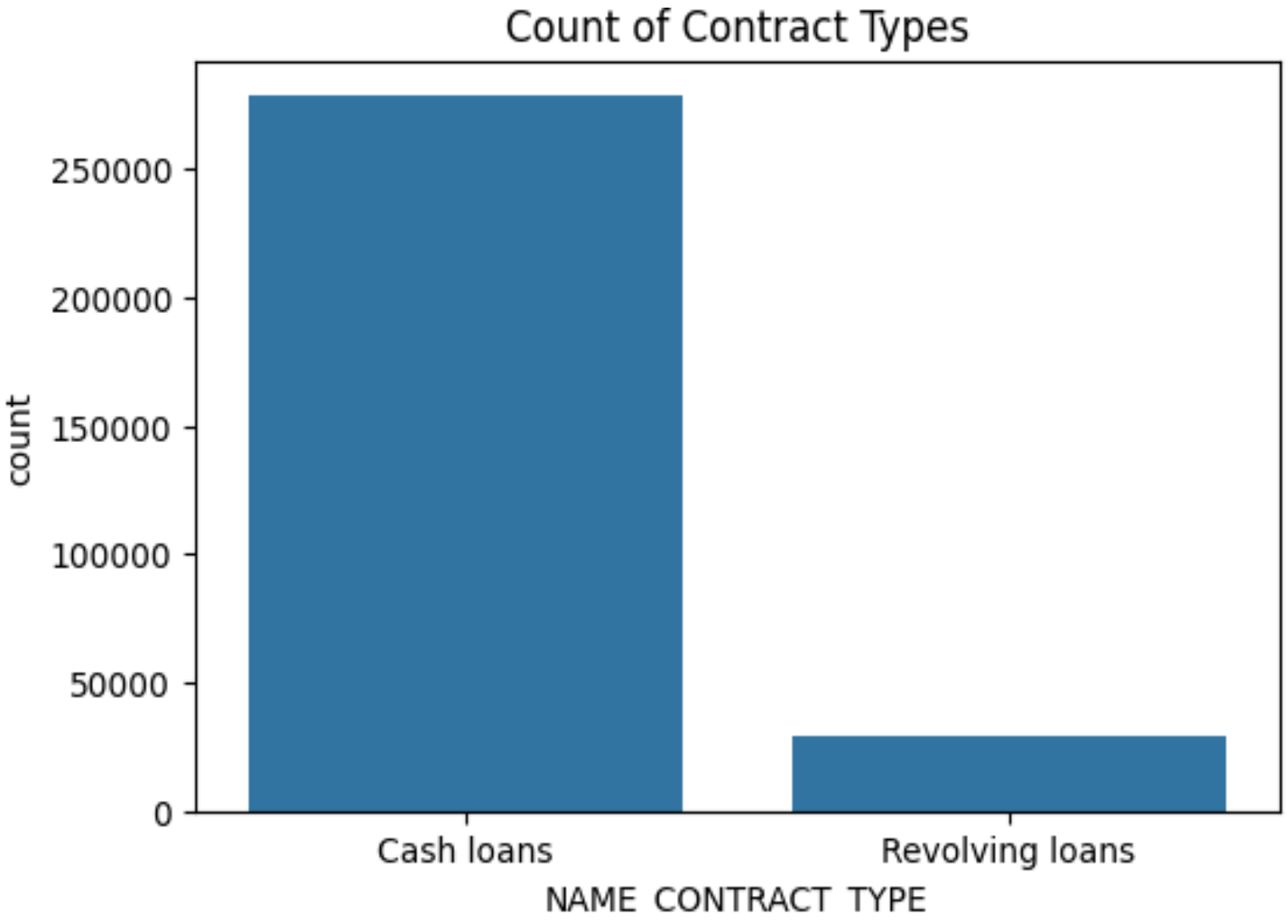
•Most clients have incomes concentrated within a specific range, with fewer high-income outliers.

•Income is a crucial factor influencing credit risk.



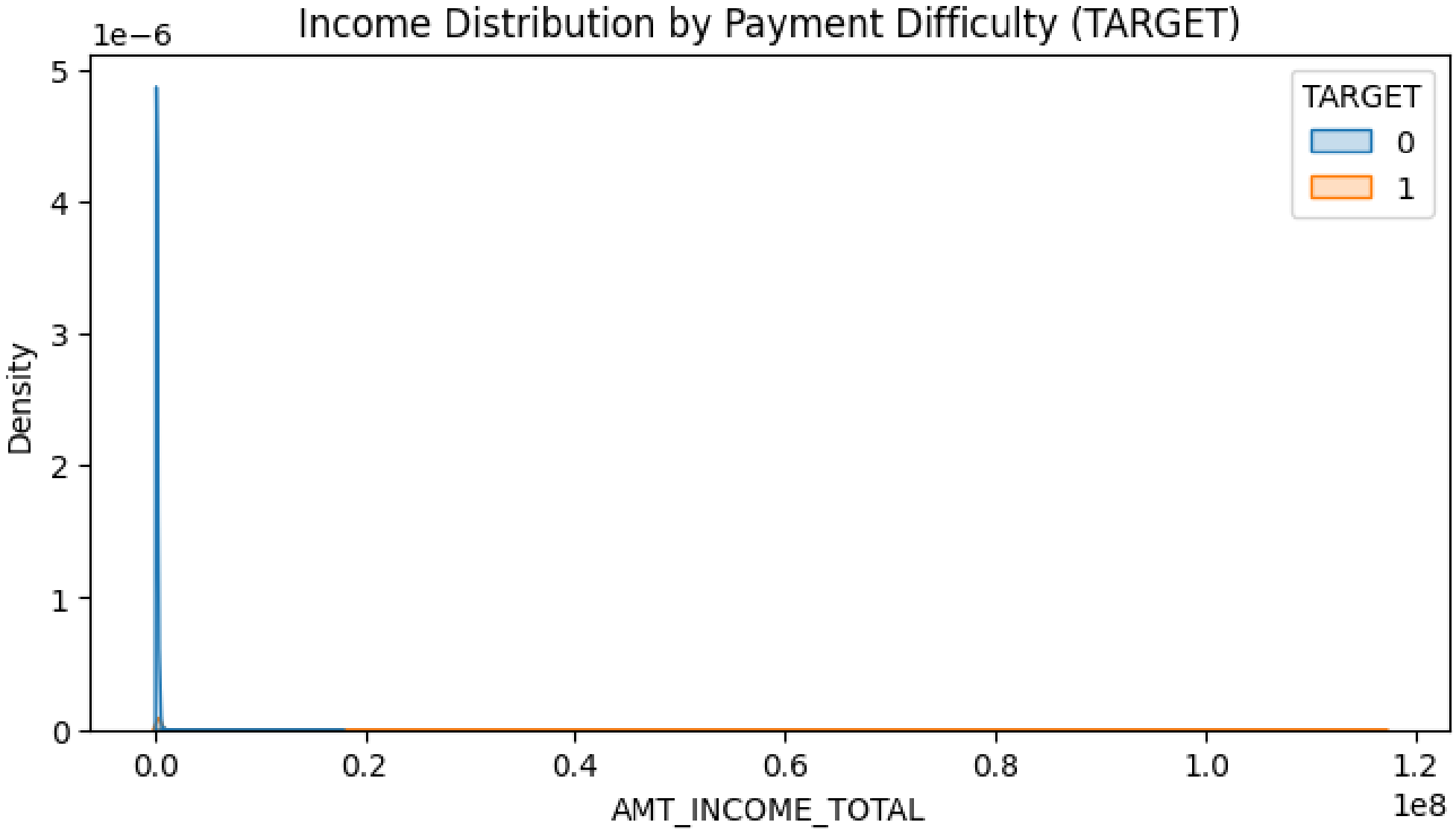
Univariate Analysis:-

•Illustrates the frequency of different loan contract types applied for.
•Helps understand which loan types are most common and may impact risk differently.



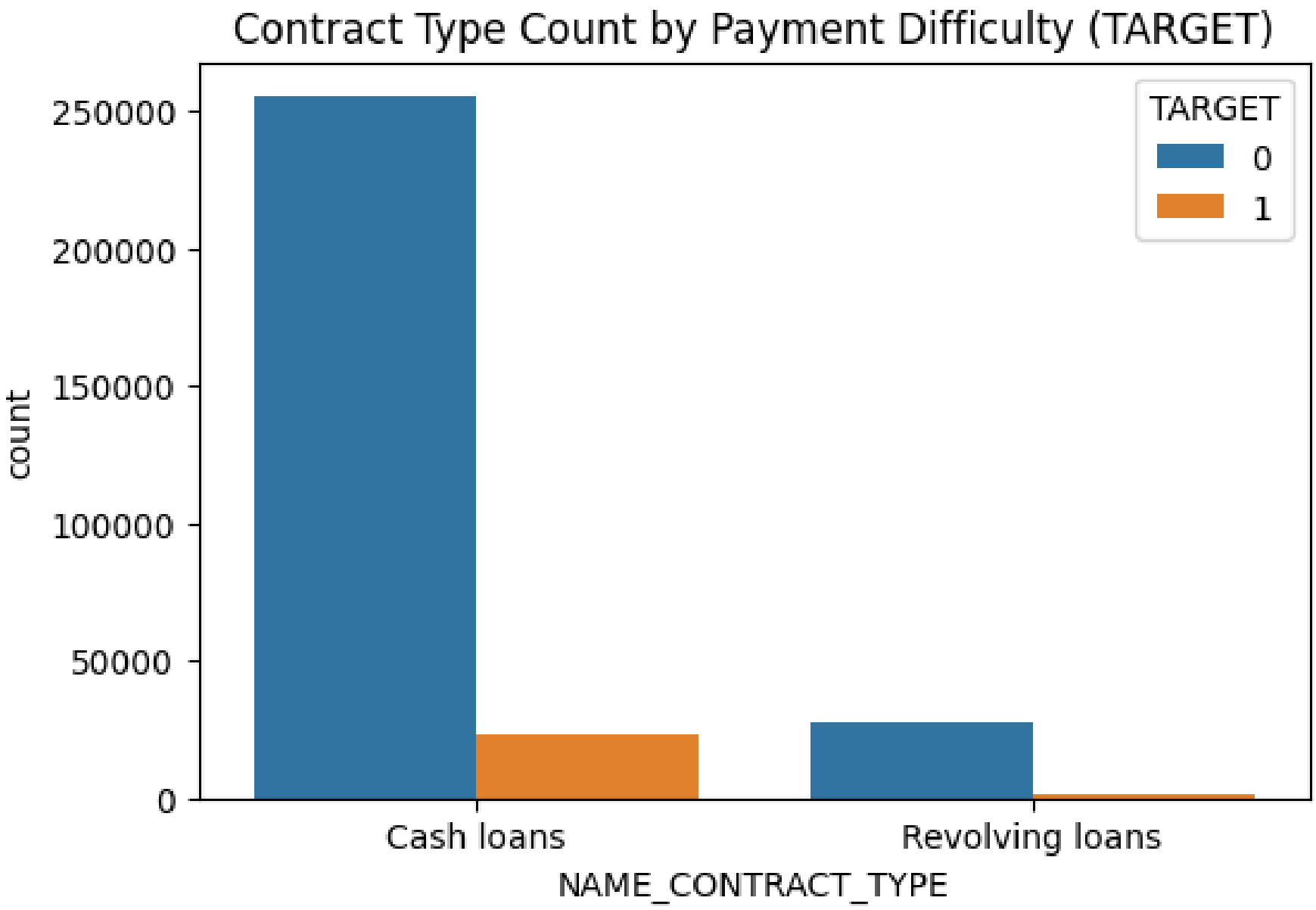
Segmented Univariate Analysis:-

- The KDE plot compares income levels of clients who repaid loans (TARGET = 0) and those who had difficulties (TARGET = 1).
- It shows that clients facing repayment difficulties generally have lower income compared to those who repay successfully.
- This suggests that lower income could be an important factor influencing the risk of default.
- Understanding this helps in assessing the creditworthiness of applicants based on income.



Segmented Univariate Analysis:-

- The count plot displays the distribution of contract types segmented by repayment status.
- It reveals that certain contract types have higher counts among clients with repayment difficulties.
- This helps identify which types of contracts are riskier and may need closer monitoring or special terms.
- Such insights guide credit policy and risk mitigation strategies.



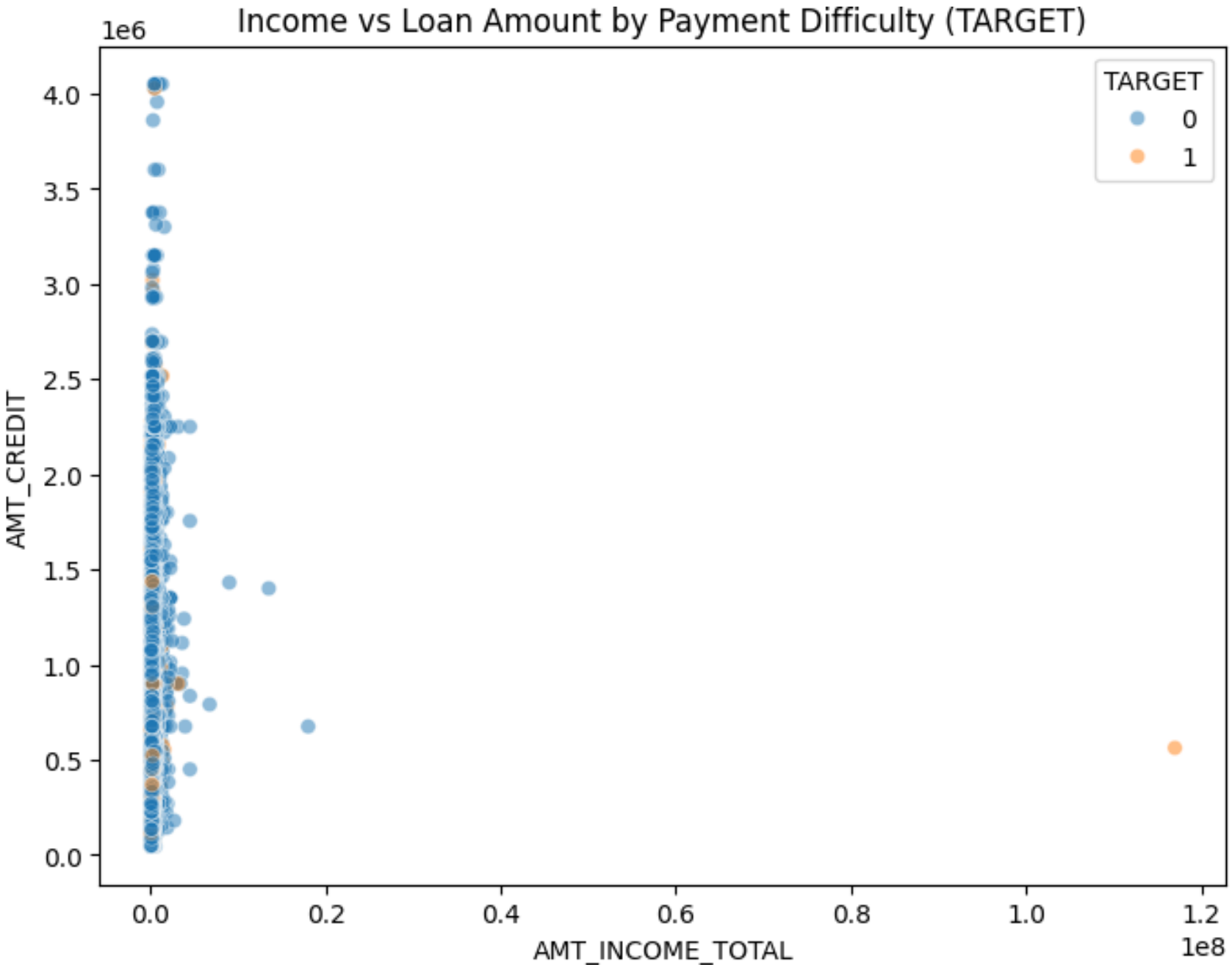
Bivariate Analysis:-

•This scatter plot shows the relationship between applicant income and the loan amount requested, segmented by payment difficulty (TARGET).

•Clients with higher incomes generally apply for higher loan amounts, indicating a positive relationship between income and credit amount.

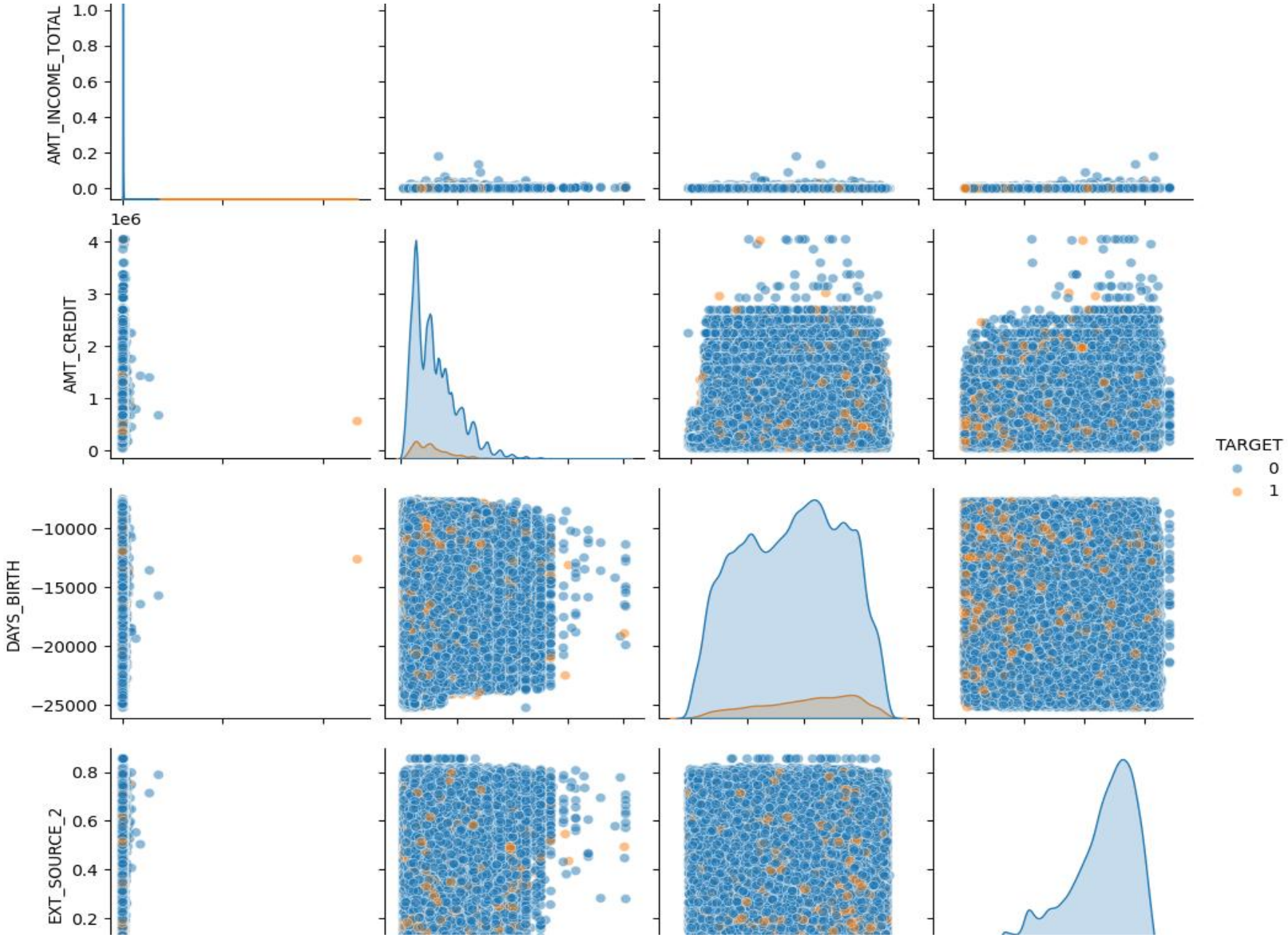
•Applicants with payment difficulties (TARGET = 1) are more concentrated in the lower income range while still taking relatively high loan amounts.

•This mismatch between income and credit amount is a strong indicator of higher default risk and is important for credit risk assessment.



Multivariate Analysis:-

clients with payment difficulties (TARGET = 1) generally have lower incomes, higher relative credit amounts, and weaker external scores compared to non-defaulters, indicating combined effects of multiple risk factors.



Multivariate Analysis:-

Note:
Due to the density of data points, some elements in the graphs may not be fully visible. However, the overall patterns and insights remain clear and are discussed accordingly.

Correlation Analysis:-

- The heatmap shows strong relationships among key financial variables for defaulters, indicating that income, credit amount, and external risk scores jointly influence the likelihood of default.
- The absence of extremely high correlations indicates low multicollinearity, meaning these variables provide distinct information about defaulters.
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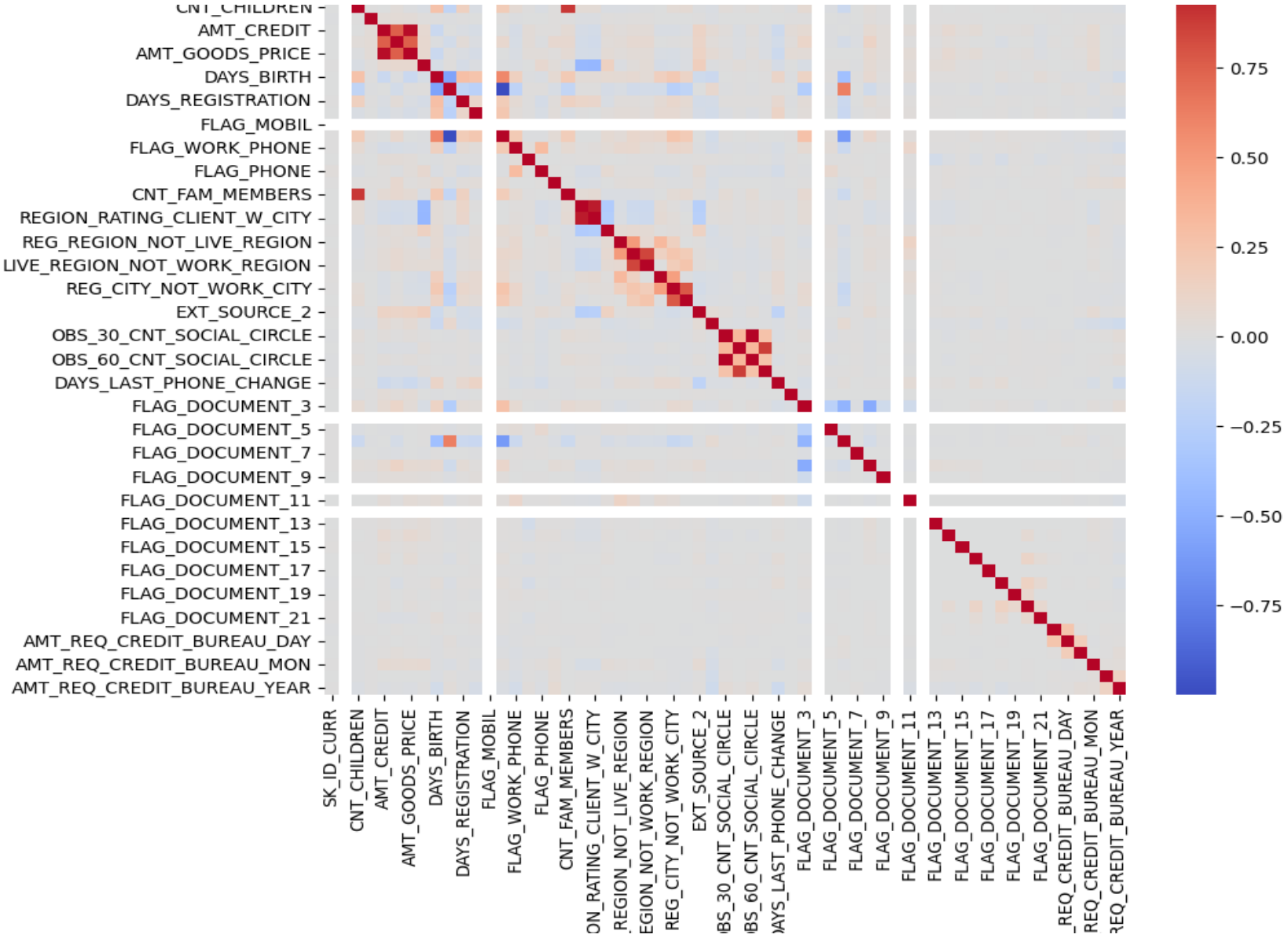
- Some financial variables show moderate correlations, suggesting that multiple factors together contribute to default risk rather than a single variable.

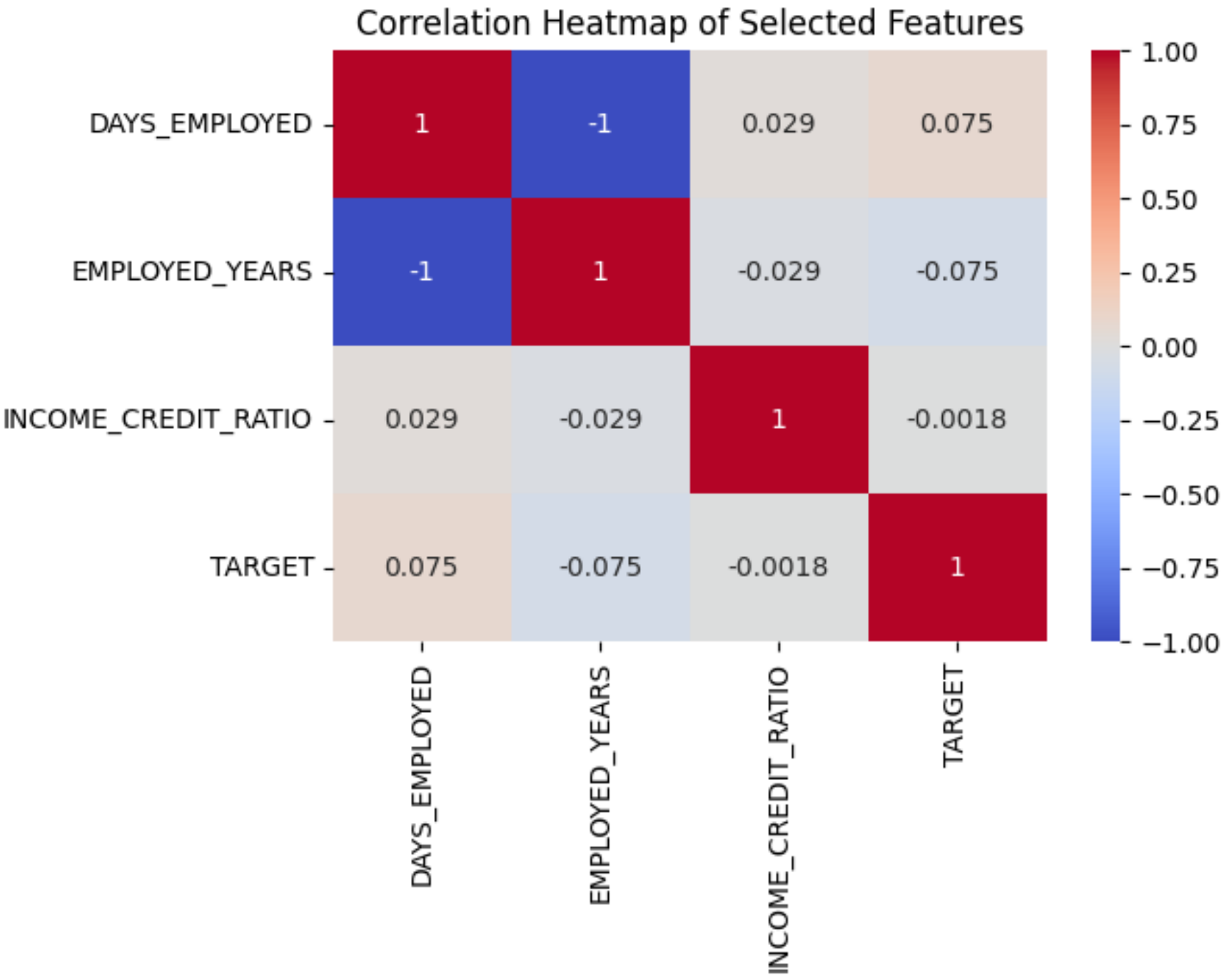


Correlation Analysis:-

Note:
Due to the density of data points, some elements in the graphs may not be fully visible. However, the overall patterns and insights remain clear and are discussed accordingly.

Feature Analysis and Selection:-

- This heatmap shows how the newly created features are related to the target variable. Features like **employment duration and income-to-credit ratio** show meaningful relationships with default risk, indicating they are useful drivers for identifying customers with payment difficulties.



Conclusion & Business Insights:-

- The dataset shows a clear imbalance, with fewer clients facing payment difficulties.
- Income, credit amount, employment duration, and external source scores are key drivers of loan default.
- Clients with lower income, higher loan amounts, and unstable employment show higher default risk.
- Correlation and multivariate analysis confirm that financial capacity strongly influences repayment behavior.
- These insights can help lenders reduce risk by adjusting loan approval criteria and interest rates.